Dr. Yixiong Jing

Research Associate, Department of Engineering, University of Cambridge
yj401@cam.ac.uk · jyx.civil@gmail.com · +44 7541 485548
Personal Webpage: https://jingyixiong.github.io/publications.html

RESEARCH PROFILE

I work at the intersection of **physical AI and infrastructure engineering**. My research develops 3D perception for civil infrastructure (large-scale point cloud segmentation, unsupervised anomaly detection), physics-grounded reconstruction and world models (differentiable simulation, neural dynamics, 3D reconstruction, and system identification), and LLM-driven embodied intelligence for construction robotics. I lead a cross-institution group of research students and engineers, with recent work in *Automation in Construction*, *Computer-Aided Civil and Infrastructure Engineering*, and at *ICRA*, *ICML*, *NeurIPS*, *and ECCV*, in collaboration with groups at **Oxford**, **TUM**, **MIT**, and **PolyU** (Hong Kong).

EDUCATION

DPhil in Engineering Science, University of Oxford, UK Jan 2021 – Feb 2025

- Thesis: Development of synthetic point cloud data and deep learning algorithms for automatic segmentation and defect detection in masonry bridges.
- Supervisors: **Prof.Sinan Acikgoz**, co-supervised by **Prof.Brian Sheil**

MEng, Kyoto University, Japan Sep 2017 – Apr 2020

BEng, Harbin Institute of Technology, China Sep 2013 – Sep 2017

EXPERIENCE

Research Associate, Department of Engineering, University of Cambridge Feb 2025 – present

- Research directions: physics-based world models and differentiable simulation; LLM-driven embodied intelligence for robot manipulation; 3D perception for infrastructure.
- Group of Cambridge Centre for Smart Infrastructure and Construction (**CSIC**) and **Laing O'Rourke Centre** led by **Prof. Brian Sheil**; researcher on **EPSRC** grant EP/Z533427/1 "**Digital Underground Construction**" (PI: B. Sheil).
- International collaborations: University of Oxford (**Active Vision Laboratory**), Technical University of Munich, MIT (**Perceptual Science Group**), PolyU (**JC STEM Lab of Machine Learning and Computer Vision**).

Head of AI, InfraMind Labs *(concurrent, part-time)* Feb 2026 – 6 Sep 2026

- Led the company's AI function: a team of 9 researchers and engineers working on automated tunnel/bridge and infrastructure inspection.
- Designed and implemented the end-to-end industrial pipeline for infrastructure point cloud inspection and monitoring.
- Inspection products deployed commercially with clients including Jacobs, National Highways, Transport for London, and Network Rail.

Research Assistant, Kyoto University, Japan Apr 2019 – Jul 2019

- Shaking table tests for the skew bridge pounding.

Junior AI developer, Zoomscape, Japan Jul 2019 – Jan 2020

- Computer vision algorithm for bolt defect detection.

PUBLICATIONS

My publications concentrate on the top destinations of ***three fields***: the leading journals of civil engineering (*Automation in Construction* and *Computer-Aided Civil and Infrastructure Engineering*) and the premier international conferences of robotics (*ICRA*), machine learning (*ICML and NeurIPS*), and computer vision (*ECCV*), reflecting a deliberate emphasis on quality over quantity. **Bold** denotes Y. Jing; † denotes corresponding author; * denotes equal contribution.

Journal articles

[J1] Wu, H., Ma, W., **Jing, Y.**†, Wang, G., Wysocki, O., & Sheil, B. (2026). Waveform unification for enhancing cross-waveform generalization in learning-based GPR inversion. *IEEE Transactions on Geoscience and Remote Sensing*. Accepted for publication.

[J2] Yang, W., Lin, W., **Jing, Y.**†, Hill, T., Wysocki, O., & Sheil, B. (2026). Tunnel Scanner: Geometry-informed synthetic point cloud generation and transfer learning for tunnel segmentation. *Automation in Construction*, 187, 106924.

[J3] **Jing, Y.**, Zhang, C., Wu, H., Wang, G., Wysocki, O., & Sheil, B. (2026). Zero-shot depth map restoration from sparse infrastructure point clouds using diffusion models and prompted segmentation. *Automation in Construction*, 184, 106839.

[J4] **Jing, Y.**, Lin, W., Sheil, B., & Acikgoz, S. (2025). 3D multimodal feature for infrastructure anomaly detection. *Automation in Construction*, *178*, 106388.

[J5] **Jing, Y.**, Zhong, J. X., Sheil, B., & Acikgoz, S. (2024). Anomaly detection of cracks in synthetic masonry arch bridge point clouds using fast point feature histograms and PatchCore. *Automation in Construction*, 168, 105766.

[J6] **Jing, Y.**, Sheil, B., & Acikgoz, S. (2024). A lightweight Transformer-based neural network for large-scale masonry arch bridge point cloud segmentation. *Computer-Aided Civil and Infrastructure Engineering*, 39, 2427–2438.

[J7] **Jing, Y.**, Sheil, B., & Acikgoz, S. (2022). Segmentation of large-scale masonry arch bridge point clouds with a synthetic simulator and the BridgeNet neural network. *Automation in Construction*, 142, 104459.

Conference proceedings

[C1] Ying, Q., Wang, G., Pan, Y., Prisacariu, V. A., Sheil, B., & **Jing, Y.**† (2026). ***NEXUS***: Neural energy fields for physically consistent contact-rich 3D object dynamics. *Advances in Neural Information Processing Systems (**NeurIPS Main Track, 2026**)*.

[C2] **Jing, Y.**, Chen, X., Wang, G., Wysocki, O., Wu, H., & Sheil, B. (2026). ***PhySPRING***: Structure-preserving reduction of physics-informed twins via GNN. *Advances in Neural Information Processing Systems (**NeurIPS Main Track, 2026**)*.

[C3] Wu, H., **Jing, Y.**†, Wang, G., Wysocki, O., & Sheil, B. (2026). **GauGal**: Gaussian-Galerkin electromagnetic inverse scattering imaging. *Advances in Neural Information Processing Systems (**NeurIPS Main Track, 2026**)*.

[C4] Wang, G., Ying, Q., **Jing, Y.**†, Wysocki, O., & Sheil, B. (2026). **ActionReasoning**: Robot action reasoning in 3D space with LLM for robotic brick stacking. *IEEE International Conference on Robotics and Automation (**ICRA**)*.

[C5] Lin, S., Xu, H., Chen, J., Li, Z., Wang, G., **Jing, Y.**, Xu, S., Zhao, R., Sheil, B., Chau, L.-P., & Liu, G. (2026). RoboFlow4D: A lightweight flow world model toward real-time flow-guided robotic manipulation. *Proceedings of the 43rd International Conference on Machine Learning (**ICML**)*, PMLR.

[C6] Wang, Y., Wang, F., Gyawali, P., Xu, Z., Klimkowska, A., **Jing, Y.**, Yang, W., Biljecki, F., Holst, C., Busam, B., Sheil, B., & Wysocki, O. (2026). UnderOneFacade: Worldwide facade semantic segmentation benchmark dataset. *European Conference on Computer Vision (**ECCV**)*, Springer.

[C7] **Jing, Y.**, Sheil, B., & Acikgoz, S. (2023). A method to generate realistic synthetic point clouds of damaged single-span masonry arch bridges. *International Conference on Structural Analysis of Historical Constructions (SAHC)*, 436–448, Springer.

[C8] **Jing, Y.**, Sheil, B., & Acikgoz, S. (2023). Extraction of key geometric parameters from segmented masonry arch bridge point clouds. *5th Joint International Symposium on Deformation Monitoring (JISDM 2022)*, 185–190.

Preprints

[P1] **Jing, Y.***, Wang, Q.*, Chen, L., Jiang, J., Wang, G.†, Wu, H., Wysocki, O., Ma, W., & Sheil, B. (2026). **BendTwin**: Robust dense-to-sparse physical reconstruction with bending-aware differentiable spring-mass models. arXiv:2608.06164. *NeurIPS 2026 Workshop PhysUnderstand* (**Oral**, non-archival). Under review at *IEEE International Conference on Robotics and Automation (ICRA)*.

[P2] Li, X.[*], **Jing, Y.**[*], Ding, J., Li, W., Busam, B., Wang, G., & Sheil, B. (2026). Track2Art: Articulated object model recovery with visual-geometric track representations. arXiv:2609.27675. Under review at *IEEE International Conference on Robotics and Automation (ICRA)*.

[P3] Wang, G.[*], Ye, P.[*], Ying, Q., **Jing, Y.**[†], Ma, Y., Chen, H., Wu, H., Wysocki, O., Duan, M., & Sheil, B. (2026). A brain-inspired hierarchical framework for zero-shot robot task reasoning and execution. arXiv:2609.05985. Submitted to *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*.

[P4] Ye, P., Ma, Y., Chen, H., Wang, G., **Jing, Y.**, Sheil, B., Du, Y., & Adelson, E. (2026). You Only Touch Once: 6-DoF object pose estimation from single tactile contact. arXiv:2606.28899.

[P5] Wang, G., Zhang, X., Xin, Y., Ma, W., Liu, J., Ma, Y., Ingham, J., Wu, H., **Jing, Y.**, Wysocki, O., & Sheil, B. (2026). SwingRL: Adaptive observation reinforcement learning with world-model prediction for cable-suspended hoisting control. arXiv:2609.33053. Under review at *IEEE Robotics and Automation Letters (RA-L)*.

[P6] Wang, G., Liu, Z., Wu, H., Ma, W., **Jing, Y.**, Wysocki, O., & Sheil, B. (2026). Physical SLAM: Physics-aware Simultaneous Localisation and Mapping. arXiv:2610.17260. Under review at *IEEE/ASME Transactions on Mechatronics*.

Under review

[U1] **Jing, Y.**[*], Sun, Z.[*], Ying, Q., Wang, G.[†], & Sheil, B. (2026). PNEO: Pairwise neural energy operators for scalable collision dynamics of 3D deformable objects. Under review at International Conference on Learning Representations (*ICLR*).

[U2] Wu, H., **Jing, Y.**[†], Deng, S., Wang, G., Wysocki, O., & Sheil, B. (2026). Neural Maxwell Projector: Physics-informed domain adaptation for cross-waveform electromagnetic projection. Under review.

[U3] Wu, H., **Jing, Y.**[†], Wang, G., Wysocki, O., & Sheil, B. (2026). Re-excitable response representations from a single-waveform radar observation. Under review at International Conference on Learning Representations (*ICLR*).

[U4] Lehmberg, D., Duhr, F., Deryło, A., Wang, F., Zhang, Q., Li, W., **Jing, Y.**, Yang, W., Yang, J., Jutzi, B., Sheil, B., Busam, B., Xia, Y., & Wysocki, O. (2026). JAB: Injecting geometric features into point cloud neural networks. Under review at *3DV*.

[U5] Chai, B., Zhang, T., Wu, S., Zuo, D., Feng, X., **Jing, Y.**, Wang, G., Fan, Z., & Zou, D. (2026). EagleDepth: Efficient fine-grained monocular depth estimation via pixel diffusion decoder. Under review at International Conference on Learning Representations (*ICLR*).

[U6] Zhang, E., Dai, M., Ma, J., Liao, Y., Sun, G., **Jing, Y.**, Wang, G., Li, M., Shao, L., & Wang, H. (2026). Learning physically grounded latent actions from human and robot videos. Under review at International Conference on Learning Representations (*ICLR*).

INVITED TALKS & PRESENTATIONS

- Invited lecture, Dongtumuwu Salon, Southeast University, Nanjing, China: "**From 3D Perception to Physics-Aware Embodied AI for Construction**", 14 July 2026.
- Invited lecture, JC STEM Lab of Machine Learning and Computer Vision, PolyU, HK, China: "**Physics-Driven 3D Reconstruction and Generation for Embodied AI**", 31 July 2026.

PROFESSIONAL SERVICE

- Journal reviewer: *Automation in Construction; Underground Space; ISPRS Journal of Photogrammetry and Remote Sensing; Journal of Computing in Civil Engineering; Engineering Structures*.
- Conference reviewer: European Conference on Computer Vision (ECCV) Workshop; International Conference on Learning Representations (ICLR).

TECHNICAL SKILLS & LANGUAGES

- Methods: 3D vision and point cloud learning; differentiable physics and simulation; generative models; LLM/VLM-driven robot planning; zero-/few-shot learning.
- Tools: PyTorch, PyTorch3D, Open3D, ROS, CUDA, physical and robot simulators (e.g., MuJoCo, Pybullet, COSMOS).
- Languages: Chinese (native); English (fluent); Japanese (basic).